8 NMAP TO DISCOVER LIVE HOST USING NMAP SCANS (ARP, ICMP, TCP)

AIM:

To study the working of the Nmap tool and discover live host in a network using different host discovery techniques such as ARP Scanning, ICMP Scanning, TCP Scanning, and DCP, UDP on the toyhackme platform.

STEPS:

1. Open the toyHackme platform and select the Nmap room.
2. Study the network topology and identify the source and destination host
3. Use ARP request to discover the MAC addresses of device available in the local network.
4. Observe how ARP Packets are broadcast within the network and identify the devices that recieves the request
5. Use the ICMP Protocal to send Ping request between 2 systems and observes the responses
6. Analyzes the packet flow and identity the exchange of ARP Packets before ICMP communication
7. Perform ARP host discovery using the Nmap command identify all active host in the local network
8. Perform ICMP Address Echo Scanning and analyze the response generated by

9. Perform ICMP [illegible] analyze the response generated by the target system
10. Perform ICMP Address Mask scanning and compare the results with previous scans
11. Perform TCP SYN scanning to identify live hosts by analyzing SYN and ACK packets
12. Perform UDP scanning and observe the how Nmap identifies active hosts through UDP packets.
13. Perform a reverses DNS lookup to obtain the hostnames of the discovered systems.
14. Analyze the advantages and limitation of each scanning method.

RESULT:-

Thus live host networks were successfully identified using different Nmap scanning techniques. ARP was found to be effective for discovering devices in the local network, while ICMP, TCP and UDP scanning techniques were useful for identifying active host and verifying.